

# Lab 02

## Cleaning water and air quality data

Teaching Week 02 | Individual work. Submit in R or Python.

### Purpose

Environmental measurements may contain missing values, inconsistent units, and results below a reporting limit. Build an ordered cleaning script, preserve the original values, and show how your decisions change the summaries used to answer one descriptive question.

### Learning outcomes

After completing this lab, you should be able to

- diagnose type, unit, missingness, and reporting-limit problems;
- separate raw data from reproducible transformations;
- compare distributions before and after cleaning; and
- justify rather than conceal exclusions and imputations.

### Dataset choice

Choose one of the verified open datasets below. Each option supports the assessment when your analytical table satisfies the data contract in the next section. The expected variables describe the published record and may differ in detail from the extract you retrieve; reconcile any difference in your report.

#### Option A | Nova Scotia Grab-Sample Water Quality

**Publisher.** Nova Scotia Environment and Climate Change

**Official record.** <https://data.novascotia.ca/d/wncu-ppda>

**Record identifier.** wncu-ppda

**Licence and format.** Nova Scotia Open Government Licence; CSV

**Expected variables.** station; sample date; parameter; result; unit

**Use in this lab.** clean and transform water-quality measurements

**Why it fits.** Long-form measurements support checks of missing values, units, and reporting limits.

Filter to a manageable station, period, and set of parameters before exporting CSV. The record includes CharacteristicName, ResultValue, ResultUnit, detection conditions, and quantitation limits; the spelling of exported headings may differ. Read those fields together. One laboratory result is one row, so several rows may describe the same water sample. Do not average different chemicals or incompatible units together.

**Option B | National Air Pollution Surveillance Program**

**Publisher.** Environment and Climate Change Canada

**Official record.**

<https://open.canada.ca/data/en/dataset/1b36a356-defd-4813-acea-47bc3abd859b>

**Record identifier.** 1b36a356-defd-4813-acea-47bc3abd859b

**Licence and format.** Open Government Licence - Canada; CSV query or ZIP archive

**Expected variables.** station; date; pollutant; concentration; unit

**Use in this lab.** clean and compare air-quality observations

**Why it fits.** Station records support type repair, missingness analysis, and transformation.

Follow the record's Data Query Tool or ECCC Data Catalogue link. Select a station, pollutant, period, and sampling interval; retrieve the corresponding data dictionary and quality-code definitions. Keep one sampling interval in each comparison, and distinguish missing codes from measured zeroes.

**Option C | Canadian Air and Precipitation Monitoring Network**

**Publisher.** Environment and Climate Change Canada

**Official record.**

<https://open.canada.ca/data/en/dataset/4baa2ee4-a8aa-457a-af26-aa13e96ee2f4>

**Record identifier.** 4baa2ee4-a8aa-457a-af26-aa13e96ee2f4

**Licence and format.** Open Government Licence - Canada; ECCC Data Mart with delimited data files

**Expected variables.** site; date; chemical; concentration; precipitation

**Use in this lab.** prepare atmospheric and precipitation measurements

**Why it fits.** Repeated chemistry records support checks of units, missingness, and table shape.

Follow the record's Data Mart link and choose one measurement product, site, and period. Read its format and quality-code documentation before importing. Use the delimiter stated for that product; document any conversion to CSV. Confirm that your extract meets the observation requirement before committing to the option.

## Data contract

Choose one listed dataset and construct a cleaned observation-level table.

- Retain at least 200 observation-level records and three variables used in your question, including at least one measured quantity. Identifiers alone do not satisfy the analytical-variable requirement.
- Include at least one variable that needs missing-value, type, unit, or reporting-limit treatment.
- Identify any station, site, region, or time columns needed to interpret missingness.
- Record every transformation and preserve the raw value in the reproducible workflow.
- Preserve detection limits and quality flags. A result below a reporting limit is not an observed zero, and an unrecognized token is not automatically a missing value.
- Compare summaries before and after cleaning.
- Reuse Week 01 summaries and unusual-value checks.
- Document the official record URL, publisher, licence, retrieval steps, and access date.

**What the contract does not settle**

Meeting the contract does not establish that a transformation is scientifically justified.

## Selection checklist

Resolve every unchecked item before you begin the analysis.

- ☐ I chose one option from the dataset catalogue above.
- ☐ I recorded the official URL, publisher, licence, access date, and retrieval steps.
- ☐ I identified the observational unit, groups, units, identifiers, and missing-value codes.
- ☐ My table satisfies every rule in the data contract.
- ☐ I identified at least one consequential cleaning or transformation decision.
- ☐ I will compare summaries before and after cleaning.
- ☐ I will preserve the raw values and document every transformation.
- ☐ I will cite the dataset in the report.

## Cumulative method

This lab carries forward Week 01 observational-unit definitions, summaries, spread measures, and unusual-value checks. Your report must show how those earlier decisions support the current analysis rather than repeating them as disconnected exercises.

## Work program

### 1. Profile the raw data

Reproduce the Week 01 variable audit and summaries before changing any value.

### 2. Specify the cleaning contract

Define valid types, units, ranges, missing codes, duplicate rules, and the observational unit required by your question. Repeated station-date pairs are not necessarily duplicates; parameter, depth, sample fraction, and laboratory identifiers may distinguish them. Use the publisher's documentation to decide.

### 3. Implement transformations

Write one ordered script that preserves raw inputs and records counts affected by each decision. Use the Class B cleaning example as a pattern. Document any type repair or unit conversion, including the conversion formula and source. Reshape only if your question needs a different table layout.

### 4. Test the consequences

Compare non-missing counts, missingness, and the Week 01 summaries before and after cleaning, within comparable measurement units. Reuse the histogram or boxplot taught in Week 01 for one consequential change. Investigate whether exclusions are concentrated by site, time, or another meaningful group. Keep censored results distinguishable and explain which summaries exclude them.

## 5. Report limitations

Explain one cleaning choice that could change the conclusion and one additional fact you would need to resolve it. No imputation, sampling, PCA, predictive model, or significance test is required; use a method only when it answers your question and you can justify its assumptions.

## Reproducible workflow

You may complete and submit the lab in either R or Python. Both paths are marked against the same evidence and rubric, and both must run end to end from one documented entry point in your portfolio repository.

Start with the imported values and make a separate analytical table. Replace the filename and column names with those in your extract. Reading columns as text preserves entries such as  $<0.1$  until you have checked the data dictionary and chosen a documented treatment.

```
data_path <- "data/observations.csv"
raw <- read.csv(data_path, colClasses = "character",
               check.names = FALSE, na.strings = NULL)
cleaned <- raw
measurement_col <- "result"
group_col <- "station"
dir.create("results", showWarnings = FALSE)
```

**Listing 1.** Read and preserve the retrieved table in R.

```
from pathlib import Path
import pandas as pd

data_path = "data/observations.csv"
raw = pd.read_csv(data_path, dtype="string", keep_default_na=False)
cleaned = raw.copy()
measurement_col = "result"
group_col = "station"
Path("results").mkdir(exist_ok=True)
```

**Listing 2.** Read and preserve the retrieved table in Python.

Apply the documented type and value conversions to `cleaned`, leaving `raw` unchanged. These calls retain missing-value tokens as text; convert them using the publisher's definitions. Retain the downloaded file as the original reference. Continue with the current class's worked methods and write the tables and figures cited in your report to `results/`.

Run from your lab directory using `Rscript lab-02.R` or `python lab-02.py`. For a notebook, restart the session and run all cells in order. State the command or notebook procedure in your README.

Downloaded data and generated results are not course source and are not versioned in this repository. Keep the evidence the report cites in your private work repository until the deadline. Publish it in your public GitHub portfolio repository only when the instructor opens sharing. Link rather than redistribute source data unless its licence and size permit it.

## Submission package

Keep the lab private before its submission deadline, including cloud notebooks and answer-revealing links. A branch in a public repository is not private. Give the instructor access to your private work repository. By 23:59 two calendar days after the integrated lab, record the full commit SHA and its commit URL in `labs/lab-XX/README.md`, then commit and push that record. The marker reads that README on the default branch and assesses the snapshot it names. The immutable commit or release URL must resolve to that exact version of `labs/lab-XX/`. A branch URL or movable tag alone is not a fixed submission. Include

- a concise PDF report that answers the analytical question, together with the completed selection checklist;
- the complete R or Python source plus the generated tables, metrics, and figures referenced in the report;
- a README carrying the submitted commit SHA and its commit URL, the official data URL, retrieval instructions, exact execution command, and environment; and
- no redistributed source data unless its licence and size permit it.

After the deadline and when the instructor opens sharing, publish the same submitted commit and its history in your public GitHub portfolio repository and present that version in class. Preserve the submitted SHA when merging. Commit regularly as you work. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded assessment history. Keep submitted commits reachable and make corrections in new commits. The pushed README record and the instructor-retained snapshot provide the independent assessment reference; Git author dates alone are not proof of timely submission.

## Assessment standard

The course uses one rubric across all integrated labs. The best ten of the twelve weekly lab results contribute 15% of the course grade.

**Table 1.** Integrated lab marking criteria.

| Criterion                          | Weight (%) |
|------------------------------------|------------|
| Dataset provenance and suitability | 15         |
| Reuse of earlier methods           | 20         |
| Current-week analytical method     | 35         |
| Interpretation and limitations     | 20         |
| Reproducibility and communication  | 10         |
| Total                              | 100        |

Academic integrity, private GitHub exceptions, private grading, and publication limits follow the course syllabus. Generative AI output is prohibited on integrated labs, so generated code, prose, analysis, or figures cannot be submitted for lab credit. Cite the dataset and every non-trivial external source; the submitted analysis must be your own work.